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# Research on Smart Tourism System Based on Artificial Intelligence

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***Abstract*—The maturity of artificial intelligence and the personalization of tourism needs are propelling the penetration of artificial intelligence into the tourism industry. In recent years, many policies in China support the construction of smart tourism, and artificial intelligence, as technical support, is a very important factor in this process. This paper firstly analyzes the application of AI in smart tourism of China. The results show that AI can improve the quality of tourism services, simplify management functions, and promote precision marketing. In addition, there are issues such as lack of planning and imperfection of functions in the application process. According to this situation, this paper builds a smart tourism system based on artificial intelligence which is composed of four subsystems for tourists, enterprises, governments and residents respectively so as to improve the functions of the existing system.**

***Keywords—artificial intelligence; smart tourism; tourism industry; design of system***

## I. INTRODUCTION

With the development of the economy and the change in people's minds, more and more people regard travel as an integral part of their life. From 2011 to 2019, China's tourism income increased year by year, as shown in Fig.1. In recent years, tourism revenue has decreased slightly due to the epidemic, but there is still a strong development trend. To meet people's demand for personalized and convenient travel, the tourism industry uses a lot of information technology. 2014 was designated as the Year of Smart Tourism under the guidance of the National Tourism Administration, which accelerates the process of integrating information technology with tourism. As a key technology for effectively processing data, using computer reasoning technology for decision supporting and problem solving, artificial intelligence plays an important role in the construction of smart tourism.

The improvement and popularization of information technology has provided abundant resources for the development of artificial intelligence. So far, artificial intelligence has crossed the gap between the laboratory and the market, becoming a new application that can replace humans to perform cumbersome operations. Many companies are investing in the AI industry, so its scale continues to grow. Data shows that the scale of China's AI industry increases from 228.8 billion yuan in 2018 to 303.1 billion yuan in 2020, as shown in Fig.2. The layout of AI enterprises in China is divided into a basic layer, a technology layer, and an application layer. Among them, the number of AI enterprises in the application layer accounts for the highest proportion, reaching 84.05%, mainly distributed in the security, finance, robotics, and home furnishing fields. The number of enterprises in the technology layer and basic layer accounts for 13.65% and 2.30% respectively, as shown in TABLE I.

This paper first introduces the application of AI in smart tourism in China. According to it, the paper designs an AI-based smart tourism system that serves tourists, enterprises, governments and residents.



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Fig. 1. 2011-2020 China's tourism revenue.

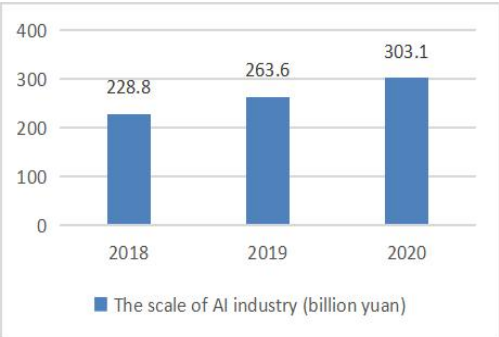


Fig. 2. 2018-2020 China's AI industry scale.

TABLE I. THE LAYOUT OF CHINA'S AI ENTERPRISES.

| Layer | Proportion | Field |
|---|---|---|
| Basic layer | 2.30% | AI chips, sensors, cloud computing, big data |
| Technology layer | 13.65% | Computer vision, voice recognition, natural language processing |
| Application layer | 84.05% | Smart security, smart finance, smart healthcare, smart robots, smart home |

## II. LITERATURE REVIEW

### A. *Smart tourism*

The concept of smart tourism comes from IBM's Smart Earth in 2008. Smart tourism is considered to be a system based on emerging technology that provides high-quality services and effectively utilizes tourism resources and even social resources which is dedicated to serving tourists, tourism enterprises, governments and destination residents[1][2].

Cloud computing, the Internet of Things, mobile terminal communication and artificial intelligence are the technological underpinnings of smart tourism. The Internet of Things refers to the technology that connects everything to the Internet via sensing devices, together with mobile communications, to get data and information about the tourism industry[3]. The role of cloud computing is to allocate information resources[4]. Artificial intelligence is a key technology in smart tourism. It helps people solve problems of tourism industry by using massive data. The four technologies of smart tourism are interrelated. The first three are basic techniques, while artificial intelligence is the core technology of smart tourism, as shown in Fig.3.

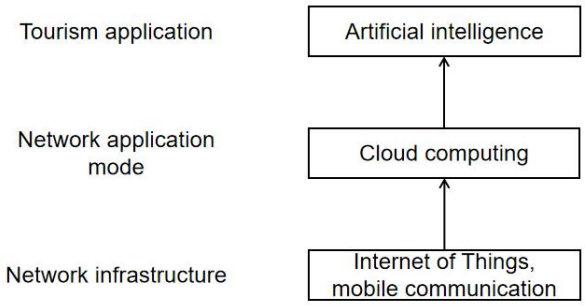


Fig. 3. Technology of smart tourism.

### B. *Artificial intelligence*

Artificial intelligence is the technology that uses machines instead of humans to realize cognition, recognition, analysis, decision-making and other functions. The essence of AI is to imitate human thinking processes in order to simplify the work of laborers[5]. AI is a system that collects information about the external environment and comes up with strategies to make intelligent behaviors with the help of the knowledge base, as shown in Fig.4.

The technologies of AI have become mature, and the research fields have also been expanding, mainly focusing on machine learning, natural language processing, image processing and human-computer interaction.



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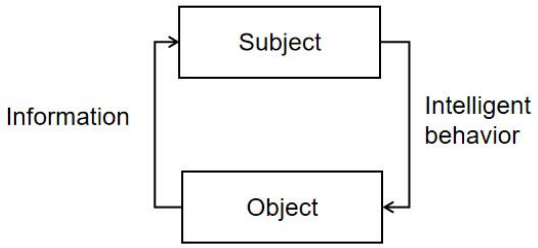


Fig. 4. The model of the AI system.

*1) Machine learning*

Machine learning is a process in which machines translate, analyze and study data with the help of algorithms to solve practical problems as well as humans[6].

*2) Natural language processing*

Natural language processing can translate human language into a form that can be processed by computers, and express computer data in human language so that humans and machines can communicate without barriers[7], which has been used in machine translation, question answering system.

*3) Image processing*

One of the common forms of information is the image. Image processing enables the computer to have visual functions[8]. It processes, analyzes, and understands images to identify different objects, and has been applied to face recognition, and intelligent security.

*4) Human-computer interaction*

Human-computer interaction is the process of exchanging information between humans and computers in a certain language[9]. Based on artificial intelligence, interaction methods such as voice-activated interaction, eye-activated interaction and virtual reality interaction have been generated.

## III. The Current Situation of the Application of AI in Smart Tourism

### *A. AI promotes the development of smart tourism*

With strong support from the government and enterprises, China has mastered the world's leading artificial intelligence technology. These technologies have accelerated change in many industries, including tourism. Tourism has gradually become an integral part of people's life. AI in the smart tourism enhances tourism service, management, and marketing.

*1) Improving the quality of tourism services*

Service quality, an intrinsic attribute of tourism, is the core competence of tourism enterprises as well as the indicator of the level of tourism development. AI permeates the whole travel process. When tourists have the intention to travel, artificial intelligence can understand their individual needs by analyzing their past browsing records and personal information. The matching destination information is then recommended to the tourists.

Tour guide service is often the most concerned factor of tourists in the process of travel. Smart tour services based on artificial intelligence not only replace manual tour guides but also allow tourists to enjoy a variety of travel services via mobile phones. The smart tour system focus on navigation and voice tour. Navigation can restore the real scene of the tourist destination to the mobile map so that tourists can easily know the related information. Voice tours can learn multiple languages and explain scenic spots instead of tour guides. It can also communicate with tourists such as recommending scenic spots[10]. In addition, the combination of AI and virtual reality technology allows visitors to interact with attractions in different dimensions.

*2) Simplifying tourism management*

The management of tourism enterprises and government is the guarantee of tourism development. The application of AI to management by tourism enterprises is most obvious in scenic spots. Scenic spots set up face recognition or two-dimensional code gate, in order to reduce labor costs. In addition, the intelligent parking system can provide managers with accurate parking data and guide visitors to empty parking lots. Prediction is one of the most important functions of AI. It combines an automatic monitoring system to analyze tourist density and tourist spatial distribution. On the one hand, it can help the scenic spot to ensure that the number of tourists is below the maximum capacity. On the other hand, the



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scenic spot can understand the operation status and predict the future number of tourists, so as to make reasonable management decisions.

Tourism authorities are also making extensive use of AI. In terms of the economy, we can understand the development of tourism in different regions through data monitoring. In terms of ecology, an emergency warning system based on artificial intelligence promotes the sustainable development of tourist destinations.

*3) Promoting precision in tourism marketing*

Marketing is essential for a company to make money. In the past, tourism marketing usually collected consumer information through market research. There were lags and singleness in data, which cannot describe consumer behavior in detail. By linking other consumer data with travel booking data, AI can help enterprises analyze consumer behavior, and push matching tourism products to tourists.

An intelligent service robot is another application form of artificial intelligence in tourism marketing. In traditional marketing, consumers will inevitably be dissatisfied with enterprises due to delayed replies. After receiving the information, the intelligent service robot breaks down the sentence structure through the algorithm, understands the problem of consumers, and provides solutions. Intelligent service robots ensure the timeliness of services.

### *B. AI brings challenges to smart tourism*

Emerging technology is a coin with two sides that has revealed many problems while enabling the rapid development of smart tourism.

*1) Lack of planning*

In response to the initiative to promote information technology in scenic spots, tourist destinations have copied the original smart tourism system and introduced artificial intelligence technology without planning based on the characteristics of tourism resources, causing a lot of waste of resources. In addition, some scenic spots invest a lot of energy at the beginning, but lack of system update and maintenance in the later stage, resulting in functions that cannot meet the needs of tourists.

*2) Imperfection of functions*

China's smart tourism system has just started. The functions of smart tourism mainly focus on precision marketing and smart tour services. And these services do not highlight the characteristics of the scenic spot. What's more, there are fewer functions that serve the government and destination residents.

## IV. CONSTRUCTION OF SMART TOURISM SYSTEM BASED ON AI

According to the literature review, the Internet of Things, mobile communication, cloud computing and artificial intelligence in the smart tourism system are interrelated rather than independent. Therefore, this paper builds a smart tourism system framework with artificial intelligence as the core technology and the rest as the foundation technologies, as shown in Fig.5. According to the service object, this system has four subsystems: smart experience system for tourists, smart operation system for enterprises, smart management system for governments, and smart service system for residents.

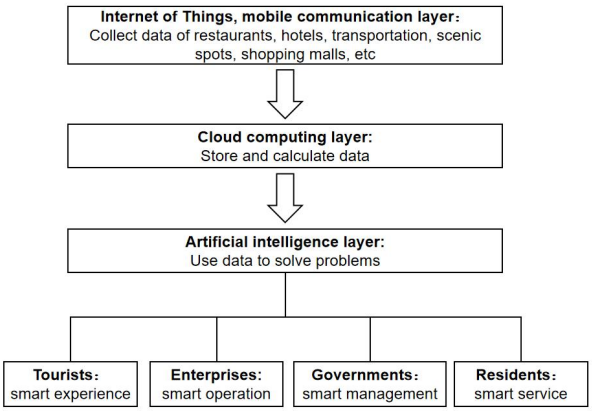


Fig. 5. The framework of smart tourism system based on AI

The first layer of the smart tourism system is IoT and mobile communication layer. IoT realizes the transmission of data between things, between things and people, and between people through sensing devices with the help of mobile communication. The second layer is the cloud computing layer, where the system stores and



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calculates the data. The third layer is the artificial intelligence layer, which can solve practical problems by processing data. It is the core layer and application layer of this system.

### A. Smart experience system

For tourists, the AI-based smart tourism system can improve the experience of the whole journey. Before the tour, it provides tourists with personalized information and helps tourists make travel plans. When tourism activities are in progress, the system provides navigation and audio guide services that break the language barrier, allowing visitors to check information on attractions anytime and anywhere. After the trip, visitors can record their journey by posting travelogues or reviews on the online platform. In addition, the smart experience system also provides online travel services for visitors to recall their trips.

### B. Smart operation system

Enterprises can use the smart operation system to analyze the relationship between consumer data and travel booking data to predict consumer behavior and conduct accurate marketing. In addition, enterprises can track consumer behavior at destinations and derive hot spots of tourists to guide their operations.

### C. Smart management system

Smart tourism provides a new management platform for governments, which realizes the collection and application of tourism industry data through flow monitoring, market supervision and public opinion forecasting. This helps governments identify problems, respond quickly and plan for the future.

### D. Smart service system

The aim of tourism development is to boost the economy of the destination. Therefore, smart tourism should not only serve tourists but also focus on the interests of the residents. The smart service system monitors and updates economic data such as tourism income and recruitment in real time, providing support for residents to make operational and management decisions. In addition, an e-commerce platform will be set up for residents to sell agricultural products and tourism products.

## V. CONCLUSION

Nowadays, tourists have higher requirements for tourism activities. The emergence of AI is driving the intelligence and convenience of tourism activities. AI, the key link of smart tourism, improve tourism services, tourism marketing and tourism management. However, artificial intelligence is a double-edged sword. While it enables the rapid development of smart tourism, it has revealed problems such as lack of planning and poor functionality. Based on this, this paper designs a smart tourism system supported by artificial intelligence for tourists, enterprises, government and residents to further improve the level of technology in service to the tourism industry.

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